Constructor University Bremen

MPFR LAB II

Part 5: Discretization Of the Controller
Spring Semester 2025

Task1.24

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Task 1.24.

Proceed as follows:

1. To start discretizing the controller first make a copy of your work up to now and rename the file. Now open the new Simulink model file.

2. Create a variable Tsample= 0.01 (in seconds) in the Matlab workspace.

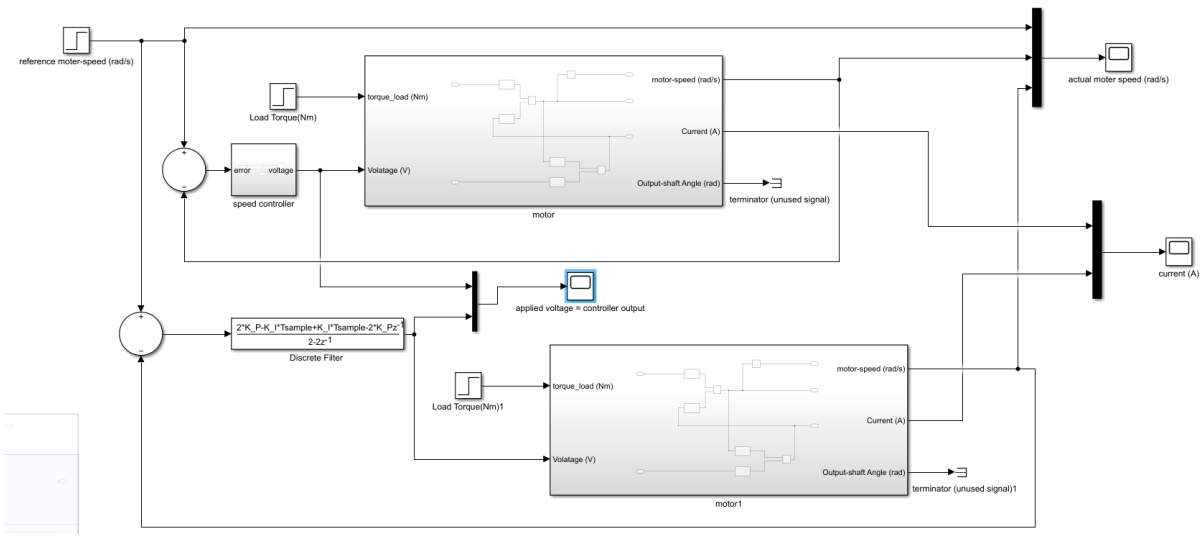
```
Tsample =

    0.0100
```

3. Now copy the motor and controller block so that you have two motors and controllers being simulated simultaneosly. We will make one of the controllers discrete. Mux their inputs/output signals before feeding them to the scopes. This way, you can compare the performance of the discrete and continuous controllers easily.

4. Now change the PID controller subsystem of one of the two controllers to that shown in Fig. 1.16. You will need the block Discrete/Discrete-Filter from the SLB. Double-click the block and specify the discrete transfer-function. Enter the numerator and denominator of G(z) from (1.44) in the expected format. In the field “Sample Time” enter Tsample.

5. Now do the simulation for a step reference-speed input and look at the response. Zoom in the voltage (controller’s output) scope – you will be able to see how the controller changes the voltage only at discrete intervals (refer to Fig. 1.17). What happens when you increase Tsample to 0.5s or higher? Also look at the current scope



Discrete filter, G(z) =

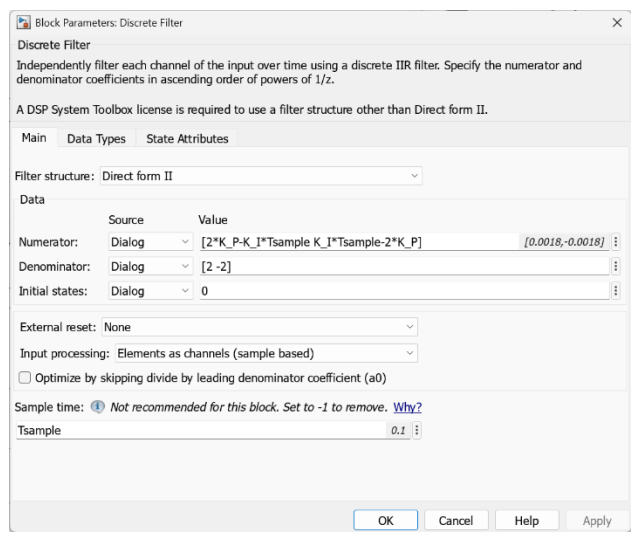
$$K_P + K_I \frac{T_P}{2}\left(\frac{1+z^{-1}}{1-z^{-1}}\right)$$

$$= \frac{K_P(2-2z^{-1}) + K_I T_P(1+z^{-1})}{2-2z^{-1}}$$

$$= \frac{2K_P - 2K_P z^{-1} + (K_I + K_I z^{-1})T_P}{2-2z^{-1}}$$

$$= \frac{2K_P - 2K_P z^{-1} + K_I T_P + K_I T_P z^{-1}}{2-2z^{-1}}$$

$$= \frac{2K_P - K_I T_P + z^{-1}(K_I T_P - 2K_P)}{2-2z^{-1}}$$




```
Tsample =

    0.1000
```